

The wonderful world of embeddings!

INTRODUCTION TO EMBEDDINGS WITH THE OPENAI API

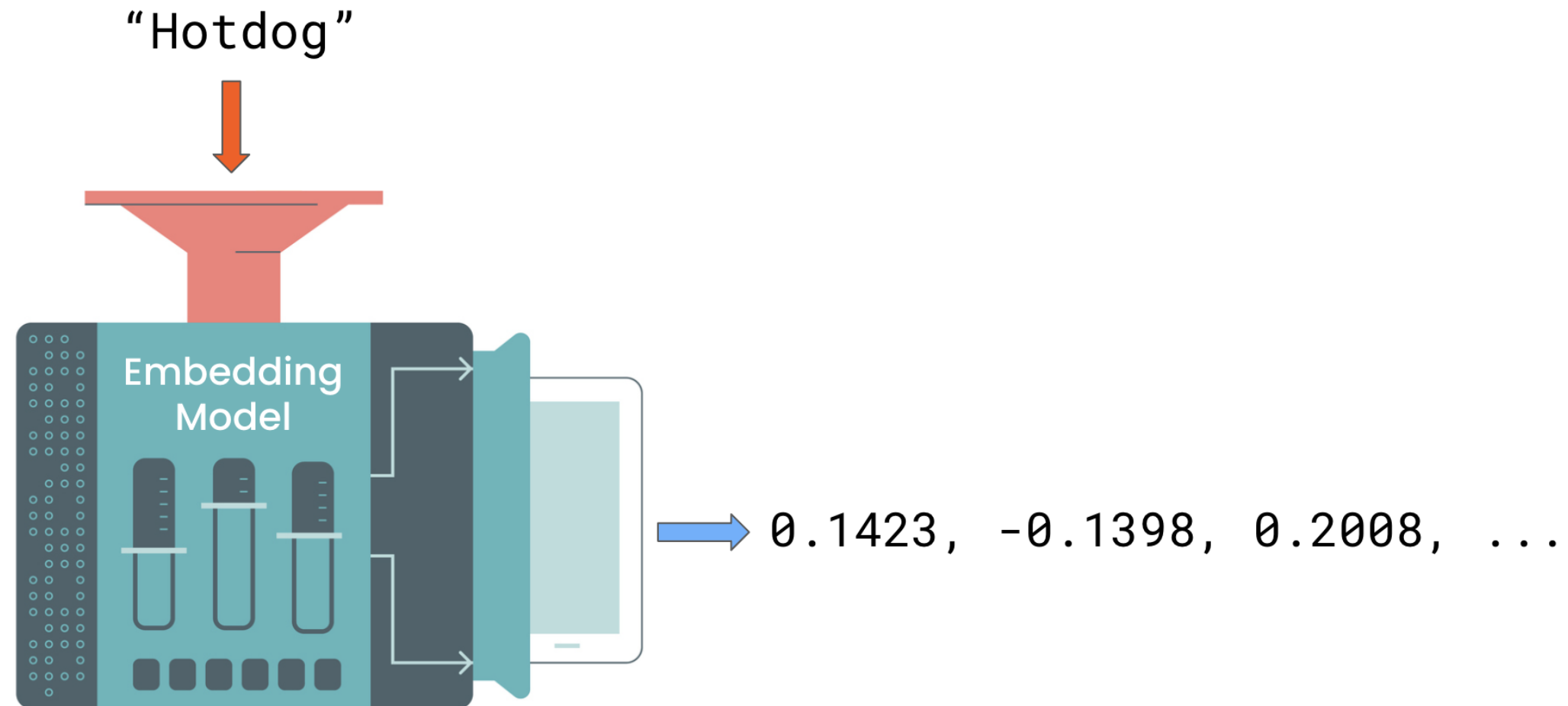


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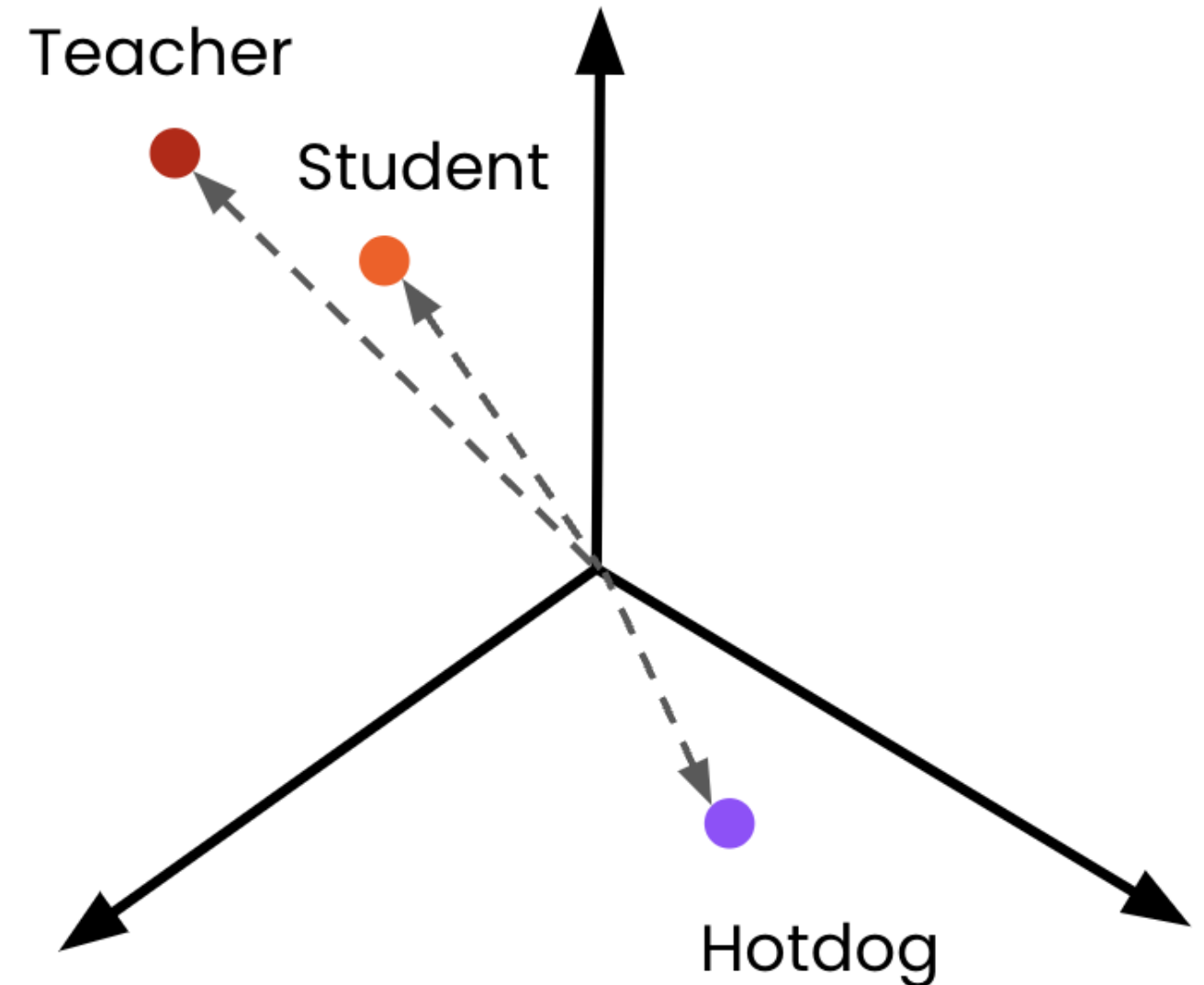
What are embeddings?

- Concept from Natural Language Processing (NLP)
- Numerical representation of text



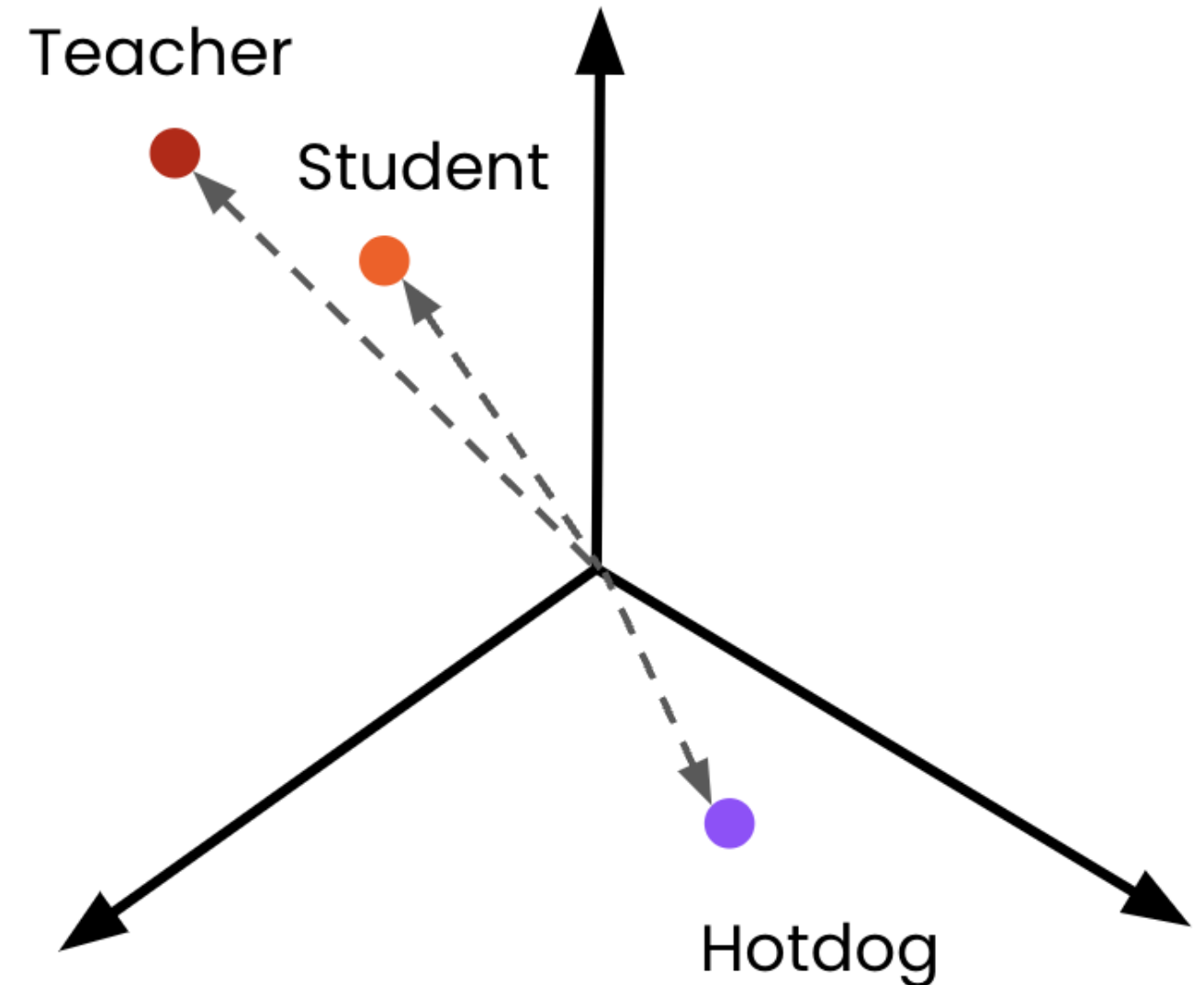
What are embeddings?

- Text is mapped onto a *multi-dimensional vector space*
vector space
- The numbers outputted by the model are the text's location in the space
- Similar words appear *closer together*
- Dissimilar words appear *further away*



Why are embeddings useful?

- Embeddings allow *semantic meaning* to be captured
- **Semantic meaning:** context and intent behind text
- **Example:**
 - "Which way is it to the supermarket?"
 - "Could I have directions to the shop?"



Semantic search engines

- Traditional search engines
 - Use **keyword** pattern matching
 - May miss the true intent
 - Will miss word variations

comfortable running shoes



[Comfortable Running Shorts](#)

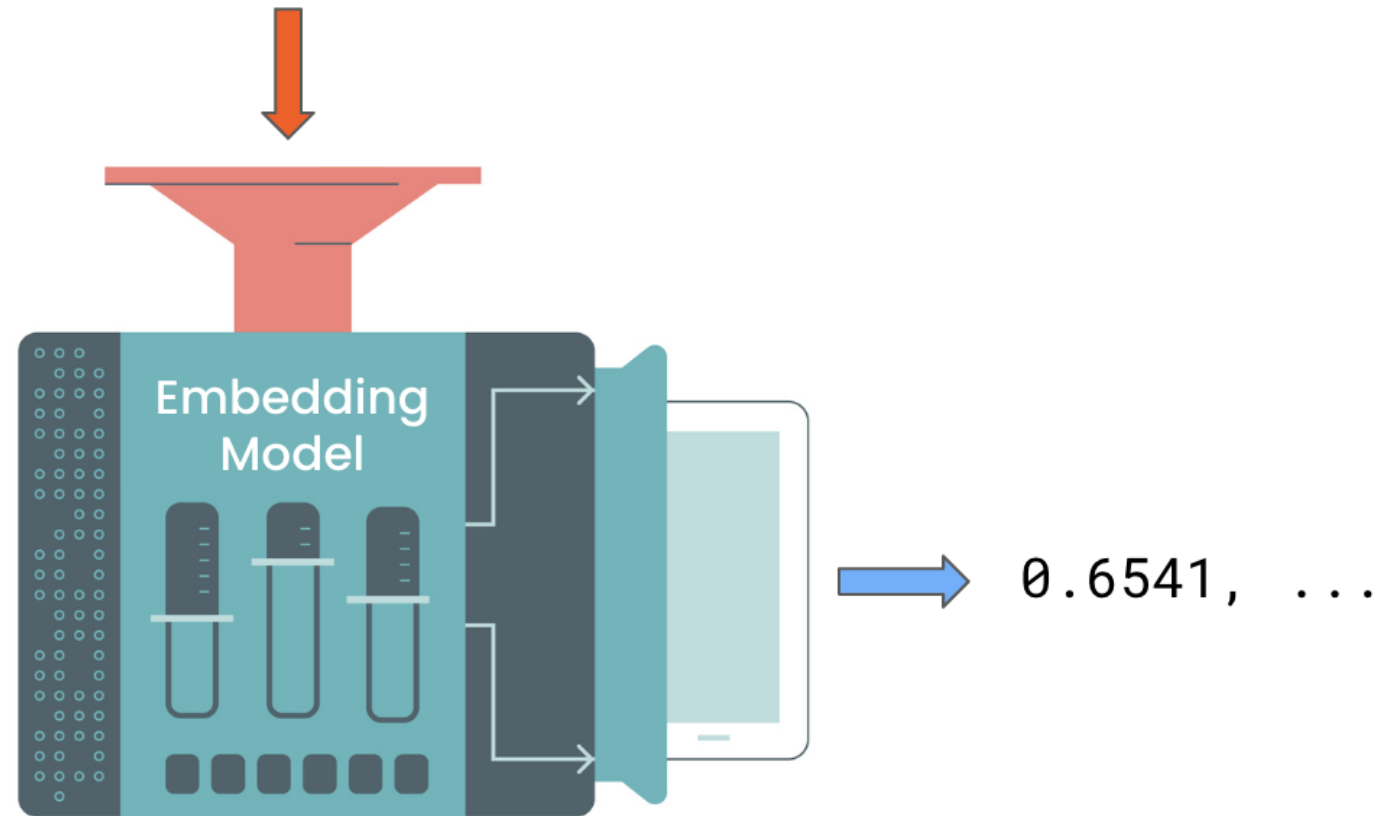
[Running Shoes for Kids - 50% Off!](#)

[Top 10 Running Routes in New York City](#)

Semantic search engines

- Use **embeddings** to understand intent and context

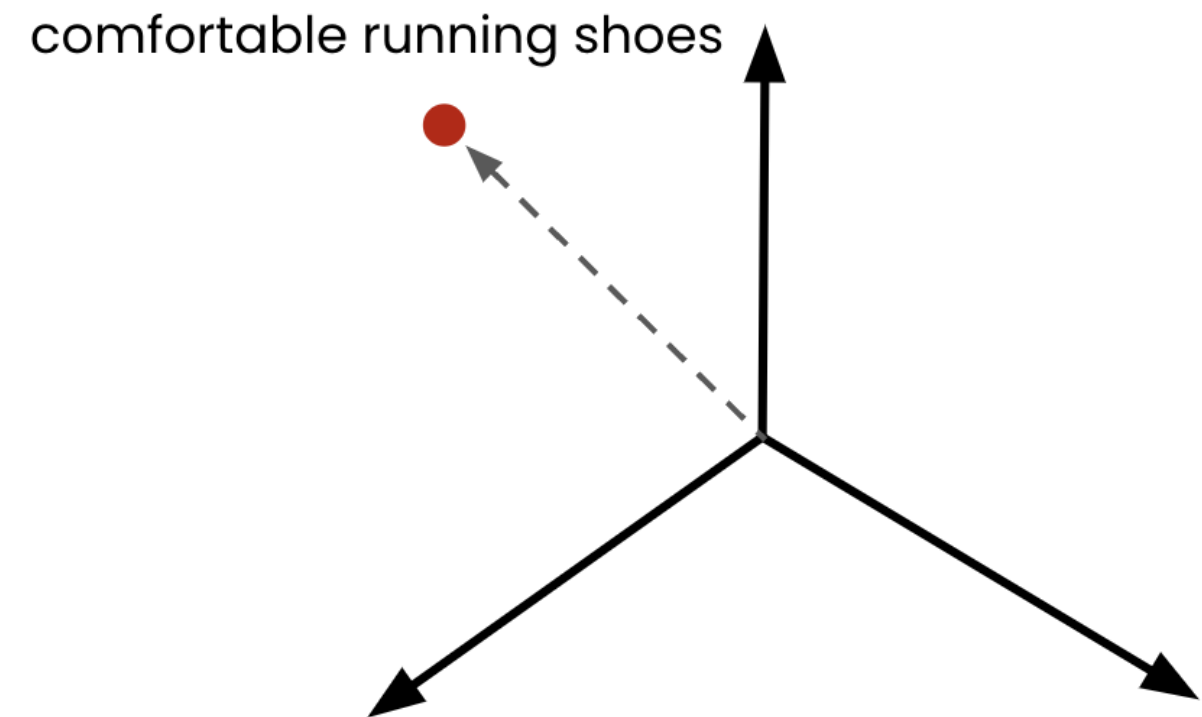
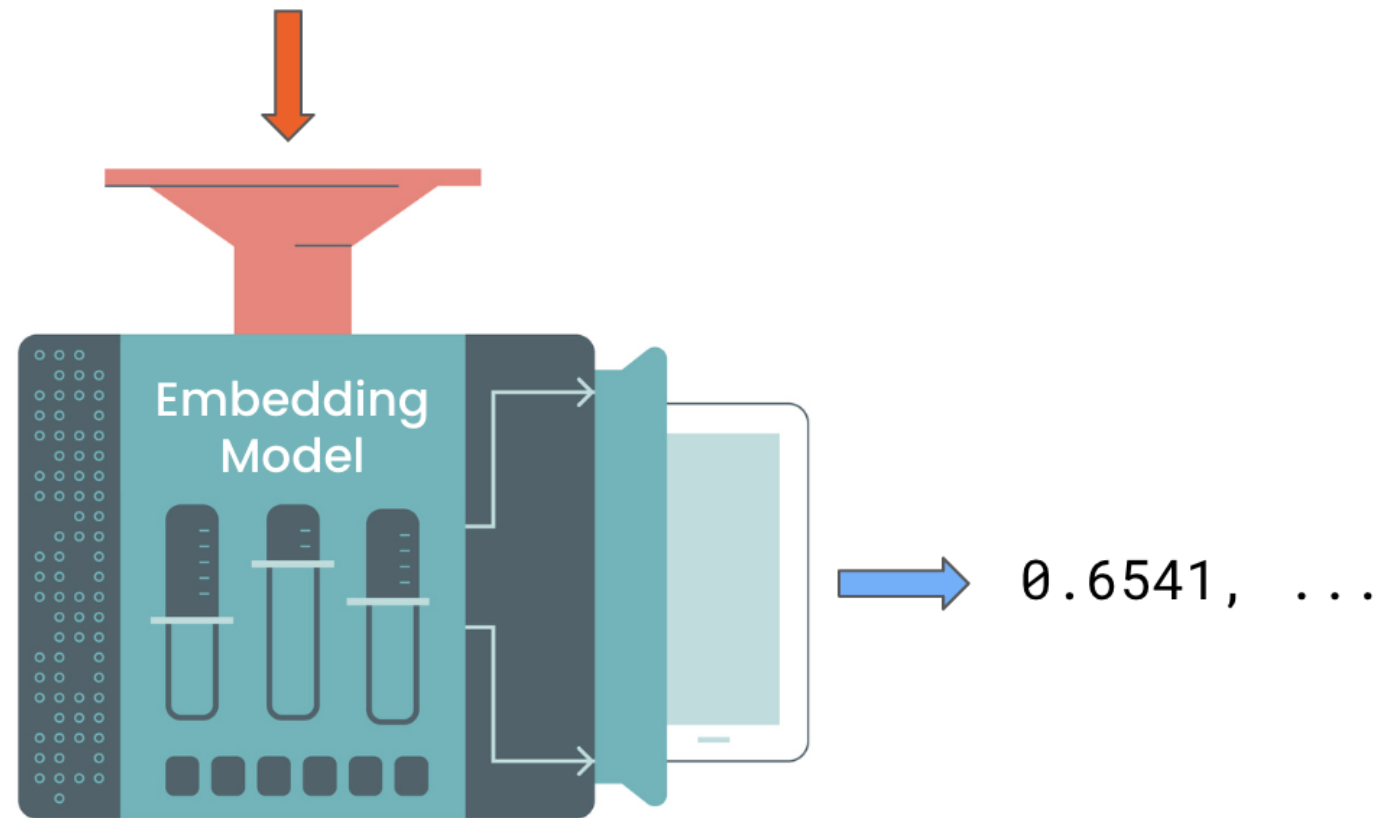
“comfortable running shoes”



Semantic search engines

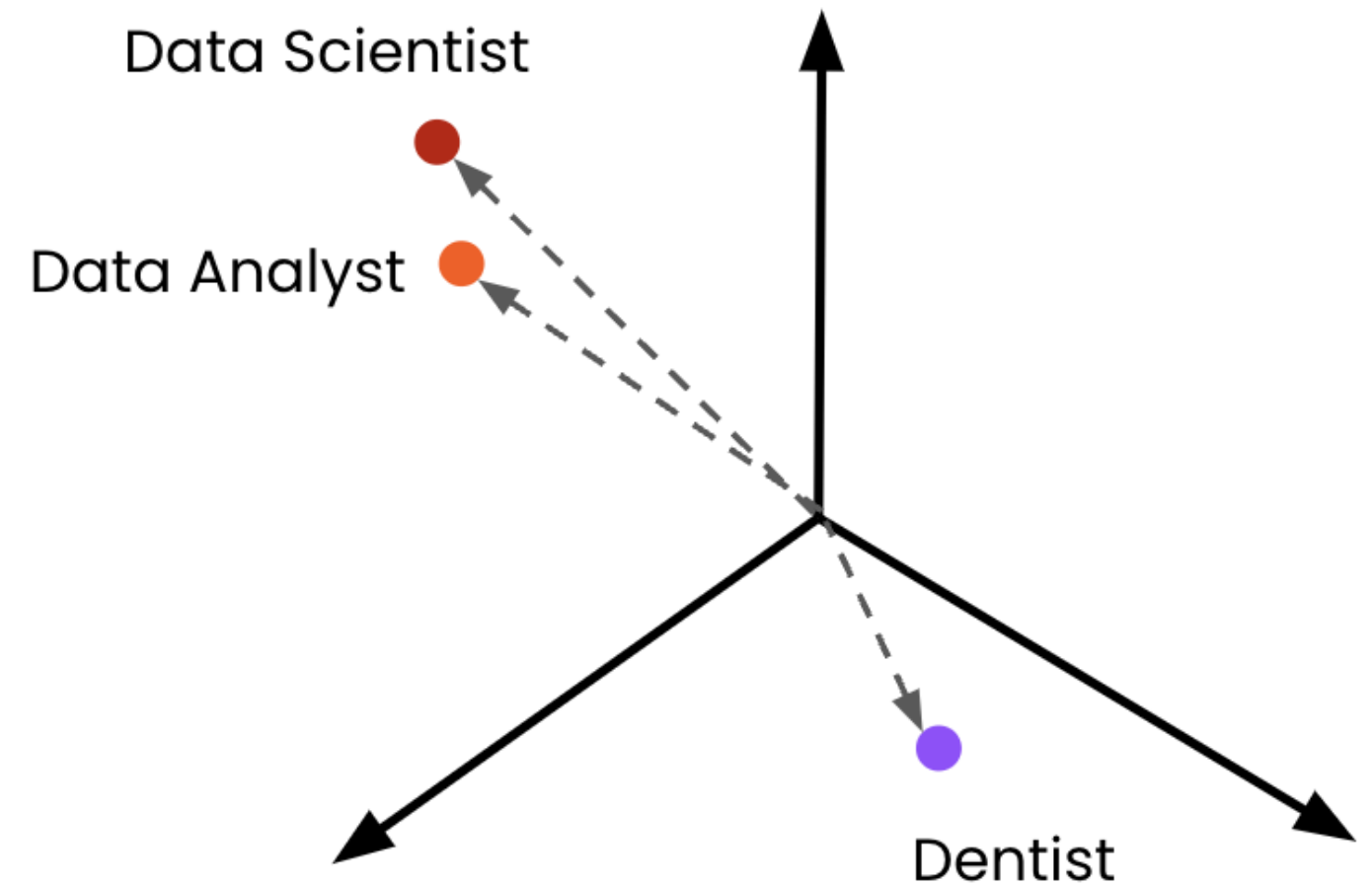
- Use **embeddings** to understand intent and context

“comfortable running shoes”



Recommendation systems

- **Example:** Job post recommendations
 - Recommend jobs based on descriptions already viewed
 - Mitigates variations in job title



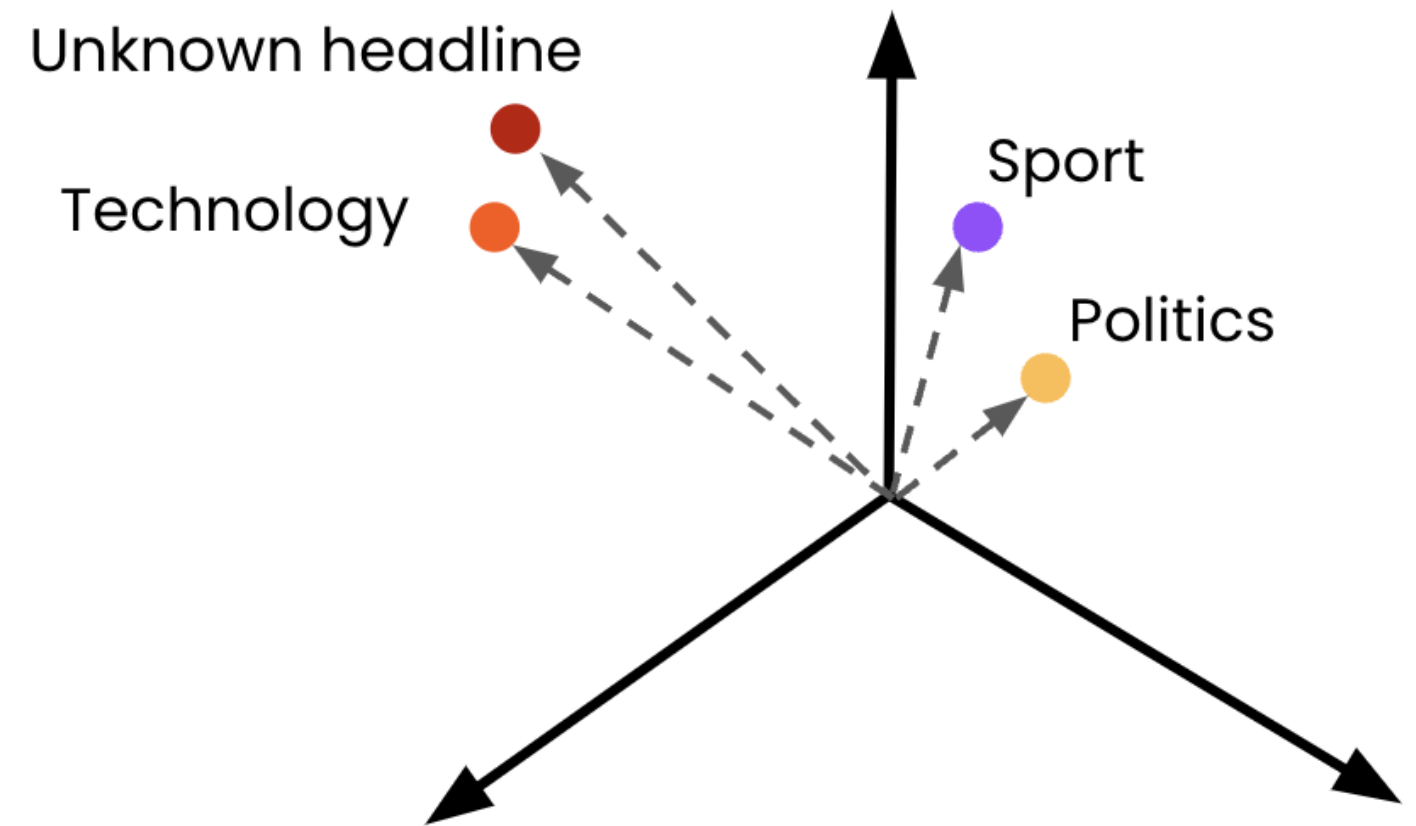
Classification

Classification tasks:

- Classify sentiment
- Cluster observations
- Categorization

Example:

- Classifying news headlines



Creating an Embeddings request

- Embeddings endpoint

```
from openai import OpenAI

client = OpenAI(api_key="<OPENAI_API_KEY>")
response = client.embeddings.create(
    model="text-embedding-3-small",
    input="Embeddings are a numerical representation of text that can be used to
measure the relatedness between two pieces of text."
)

response_dict = response.model_dump()
print(response_dict)
```

¹ <https://platform.openai.com/docs/api-reference/embeddings>

Embeddings response

```
{'object': 'list',  
  'data': [  
    {  
      "embedding": [0.0023064255, ..., -0.0028842222],  
      "index": 0,  
      "object": "embedding"  
    }  
  ],  
  'model': 'text-embedding-3-small',  
  'usage': {  
    "prompt_tokens": 24,  
    "total_tokens": 24  
  }  
}
```

Extracting the embeddings

```
print(response_dict['data'][0]['embedding'])
```

```
[0.0023064255, ..., -0.0028842222]
```

Let's practice!

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Investigating the vector space

INTRODUCTION TO EMBEDDINGS WITH THE OPENAI API



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Example: Embedding headlines

```
articles = [  
    {"headline": "Economic Growth Continues Amid Global Uncertainty", "topic": "Business"},  
    {"headline": "Interest rates fall to historic lows", "topic": "Business"},  
    {"headline": "Scientists Make Breakthrough Discovery in Renewable Energy", "topic": "Science"},  
    {"headline": "India Successfully Lands Near Moon's South Pole", "topic": "Science"},  
    {"headline": "New Particle Discovered at CERN", "topic": "Science"},  
    {"headline": "Tech Company Launches Innovative Product to Improve Online Accessibility", "topic": "Tech"},  
    {"headline": "Tech Giant Buys 49% Stake In AI Startup", "topic": "Tech"},  
    {"headline": "New Social Media Platform Has Everyone Talking!", "topic": "Tech"},  
    {"headline": "The Blues get promoted on the final day of the season!", "topic": "Sport"},  
    {"headline": "1.5 Billion Tune-in to the World Cup Final", "topic": "Sport"}  
]
```

Example: Embedding headlines

```
[
  {
    'headline': 'The Blues get promoted on the final day!',
    'topic': 'Sport',
    'embedding': [0.0015748793, ..., -0.0052598542]
  },
  {
    'headline': 'Interest rates fall to historic levels',
    'topic': 'Business',
    'embedding': [-0.030351446, ..., -0.0044289114]
  },
  ...
]
```

Embedding multiple inputs

```
headline_text = [article['headline'] for article in articles]
headline_text
```

```
["Economic Growth Continues Amid Global Uncertainty",
 ...,
 "1.5 Billion Tune-in to the World Cup Final"]
```

```
response = client.embeddings.create(
    model="text-embedding-3-small",
    input=headline_text
)
response_dict = response.model_dump()
```

- **Batching** is more efficient than using multiple API calls

```
[...]  
  
'data': [  
  {  
    "embedding": [-0.017142612487077713, ..., -0.0012911480152979493],  
    "index": 0,  
    "object": "embedding"  
  },  
  {  
    "embedding": [-0.032995883375406265, ..., -0.0028605300467461348],  
    "index": 1,  
    "object": "embedding"  
  },  
  ...  
]  
  
[...]
```

Embedding multiple inputs

```
articles = [  
    {"headline": "Economic Growth Continues Amid Global Uncertainty", "topic": "Business"},  
    ...  
]
```

```
for i, article in enumerate(articles):  
    article['embedding'] = response_dict['data'][i]['embedding']  
  
print(articles[:2])
```

```
[{'headline': 'Economic Growth Continues Amid Global Uncertainty',  
  'topic': 'Business',  
  'embedding': [-0.017142612487077713, ..., -0.0012911480152979493]}  
{'headline': 'Interest rates fall to historic lows',  
  'topic': 'Business',  
  'embedding': [-0.032995883375406265, ..., -0.0028605300467461348]}]
```

How long is the embeddings vector?

- "Economic Growth Continues Amid Global Uncertainty"

```
len(articles[0]['embedding'])
```

1536

- "Tech Company Launches Innovative Product to Improve Accessibility"

```
len(articles[5]['embedding'])
```

1536

- Always returns 1536 numbers!

Dimensionality reduction and t-SNE

- Various techniques to *reduce* the number of dimensions
- **t-SNE** (t-distributed Stochastic Neighbor Embedding)

¹ <https://www.datacamp.com/tutorial/introduction-t-sne>

Implementing t-SNE

```
from sklearn.manifold import TSNE
import numpy as np

embeddings = [article['embedding'] for article in articles]

tsne = TSNE(n_components=2, perplexity=5)
embeddings_2d = tsne.fit_transform(np.array(embeddings))
```

- `n_components` : the resulting number of dimensions
- `perplexity` : used by the algorithm, must be *less than number of data points*
- Will result in **information loss**

¹ <https://www.datacamp.com/tutorial/introduction-t-sne>

Visualizing the embeddings

```
import matplotlib.pyplot as plt

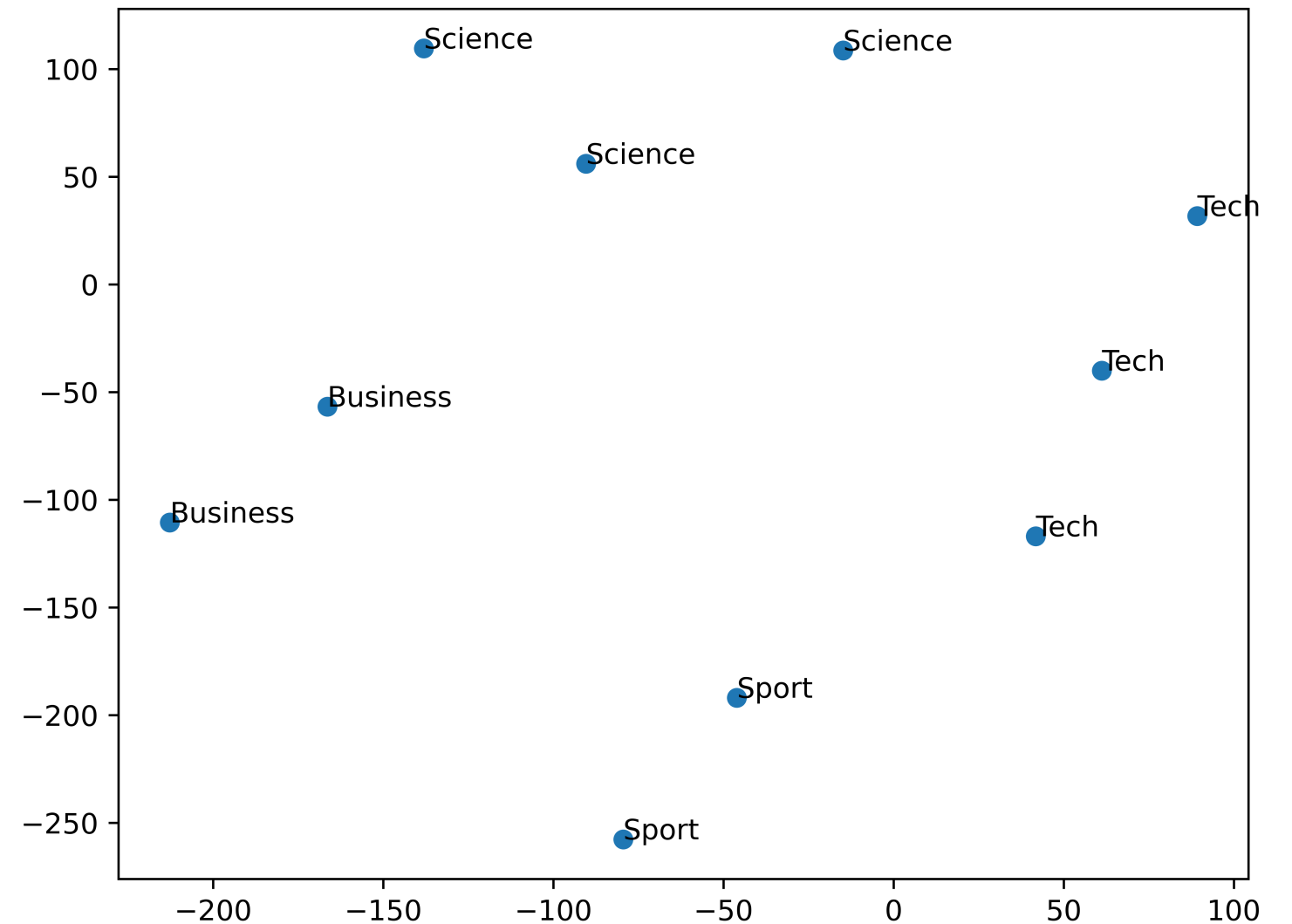
plt.scatter(embeddings_2d[:, 0], embeddings_2d[:, 1])

topics = [article['topic'] for article in articles]
for i, topic in enumerate(topics):
    plt.annotate(topic, (embeddings_2d[i, 0], embeddings_2d[i, 1]))

plt.show()
```

Visualizing the embeddings

- *Similar* articles are grouped together!
- Model captured the *semantic* meaning
- **Coming up:** Computing similarity



Let's practice!

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Text similarity

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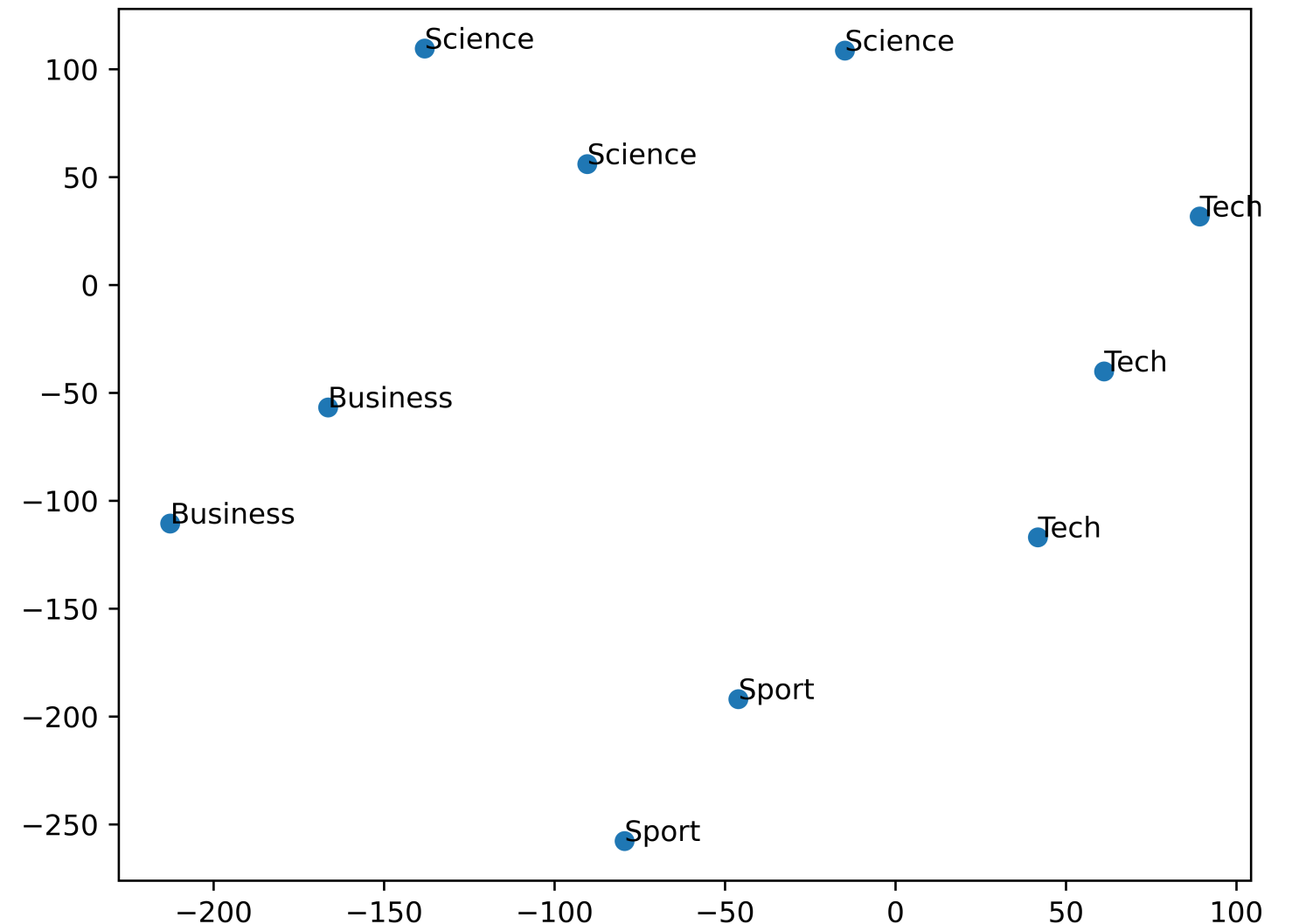


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Recap...

- Semantically similar texts are embedded *more closely* in the **vector space**
- Measuring distance allows us to measure similarity
- Enables embeddings applications:
 - Semantic search
 - Recommendations
 - Classification



Measuring similarity

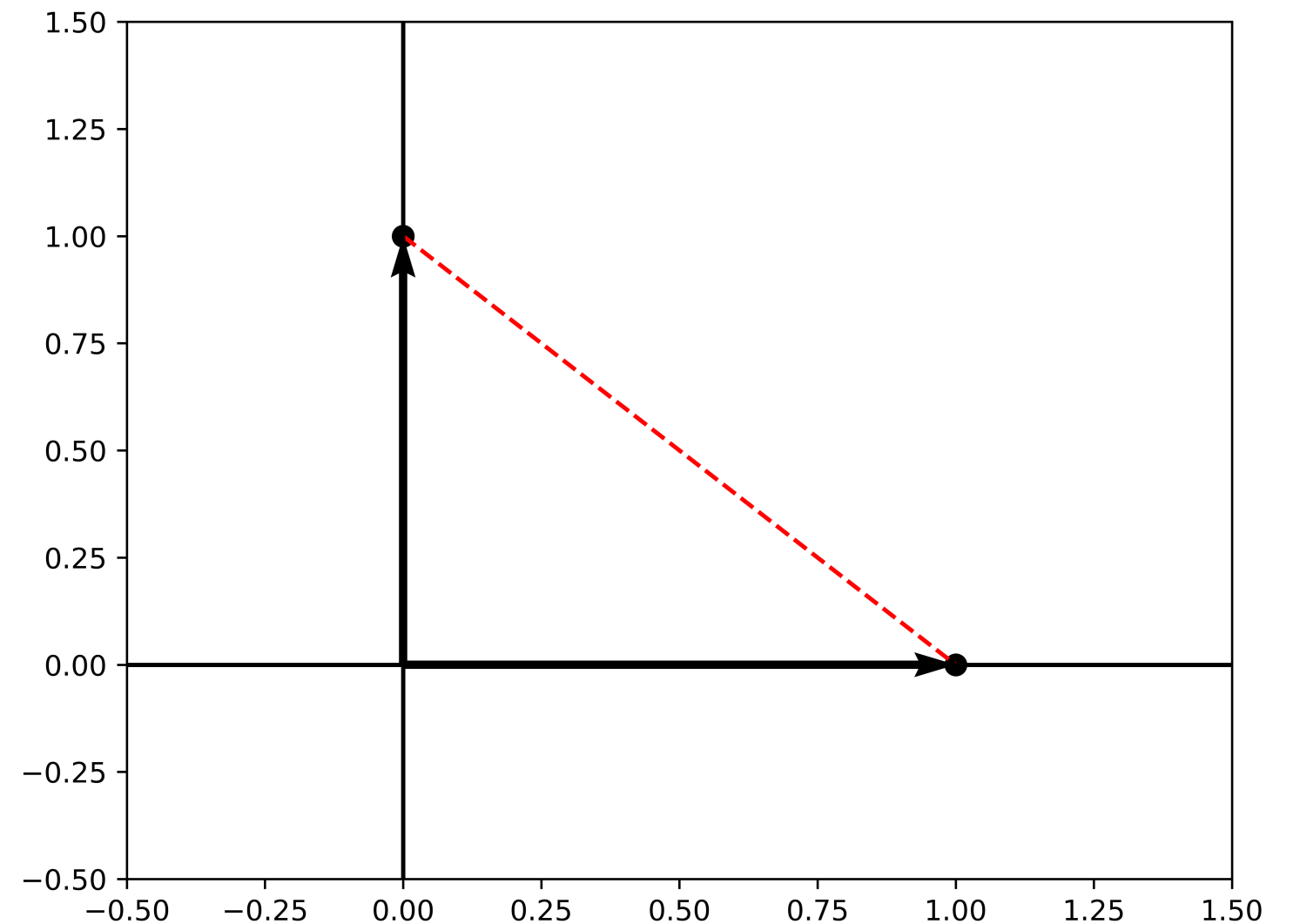
Cosine distance

```
from scipy.spatial import distance
```

```
distance.cosine([0, 1], [1, 0])
```

```
1.0
```

- Ranges from 0 to 2
- Smaller numbers = *Greater* similarity



Example: Comparing headline similarity

```
[
  {
    'headline': 'The Blues get promoted on the final day!',
    'topic': 'Sport',
    'embedding': [0.0015748793, ..., -0.0052598542]
  },
  {
    'headline': 'Interest rates fall to historic levels',
    'topic': 'Business',
    'embedding': [-0.030351446, ..., -0.0044289114]
  },
  ...
]
```

Example: Comparing headline similarity

```
def create_embeddings(texts):  
    response = client.embeddings.create(  
        model="text-embedding-3-small",  
        input=texts  
    )  
    response_dict = response.model_dump()  
  
    return [data['embedding'] for data in response_dict['data']]  
  
print(create_embeddings(["Python is the best!", "R is the best!"]))  
print(create_embeddings("DataCamp is awesome!")[0])
```

```
[[0.005056588448587894, ..., -0.04000323638319969],  
 [-0.0018890155479311943, ..., -0.04085670784115791]]  
[0.00037010075175203383, ..., -0.021759100258350372]
```

Example: Comparing headline similarity

```
from scipy.spatial import distance
import numpy as np

search_text = "computer"
search_embedding = create_embeddings(search_text)[0]

distances = []
for article in articles:
    dist = distance.cosine(search_embedding, article["embedding"])
    distances.append(dist)

min_dist_ind = np.argmin(distances)
print(articles[min_dist_ind]['headline'])
```

Tech Company Launches Innovative Product to Improve Online Accessibility

Let's practice!

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